**Chapter 8, Lecture 11**

*Marten Wegkamp*

## Efficiency

Given a sample from $N(\mu, \sigma^2)$, which estimator of $\mu$ would you prefer: $\widehat{\mu}_1 = \overline{X}_n$ or $\widehat{\mu}_2 = X_1$? Both estimates are unbiased, but $\mathrm{Var}(\widehat{\mu}_1) = \sigma^2/n <\mathrm{Var}(\widehat{\mu}_2) = \sigma^2$ for all $n \geq 2$.

Similarly, given an iid sample from a Poisson($\mu$), both $\widehat{\mu}_1 = \overline{X}_n$ and $\widehat{\mu}_2 = S^2$ are unbiased, while we saw in our simulations that

$$\frac{\mathrm{Var}(\widehat{\mu}_2)}{\mathrm{Var}(\widehat{\mu}_1)} = 3$$

The following result states that the MLE has the smallest variance among all unbiased estimates. Recall that the asymptotic variance of the MLE $\widehat{\theta}$ based on an iid sample from $f(x|\theta)$ is given by $1/\{nI(\theta)\}$.

**Theorem** *Let $X_1, \ldots, X_n$ be iid $f(x|\theta)$. Let $T_n = T_n(X_1, \ldots, X_n)$ be an unbiased estimate of $\theta$,*

$$\mathbb{E}_\theta T_n = \theta \qquad \textit{for all } \theta.$$

*Under suitable regularity conditions on $f(x|\theta)$, we have*

$$\mathrm{Var}(T_n) \geq \frac{1}{nI(\theta)}$$

This result is known as the Cramèr-Rao inequality. The lower bound is the asymptotic variance of the MLE.

*Proof:* We can assume that $\mathbb{E}[T_n^2] < \infty$ since otherwise, the left hand side is $+\infty$. Define

$$\begin{aligned} Z_n &= \frac{\partial}{\partial\theta} \log \text{lik}(\theta) \\ &= \sum_{i=1}^{n} s(X_i|\theta) \end{aligned}$$

the sum of the score functions. Recall that under suitable regularity,

$$\mathbb{E}s(X_1|\theta) = 0 \qquad \text{and} \qquad \text{var}(s(X_1|\theta)) = I(\theta)$$

This implies that

$$\mathbb{E}[Z_n] = 0 \qquad \text{and} \qquad \text{var}(Z_n) = nI(\theta)$$

whence

$$\begin{aligned} \text{Cov}(Z_n, T_n) &= \mathbb{E}[Z_n T_n] - \mathbb{E}[Z_n]\mathbb{E}[T_n] \\ &= \mathbb{E}[Z_n T_n]. \end{aligned}$$

Assume that $f(x|\theta)$ is a density of a continuous random variable. (So the expected values are integrals.) The term on the right equals 1, as the following calculation shows

$$\begin{aligned} \mathbb{E}[Z_n T_n] &= \int T_n(\mathbf{x}) \left( \frac{\partial}{\partial\theta} \log f(\mathbf{x}|\theta) \right) f(\mathbf{x}|\theta)\, \mathrm{d}\mathbf{x} \\ &= \int T_n(\mathbf{x}) \frac{\partial}{\partial\theta} f(\mathbf{x}|\theta)\, \mathrm{d}\mathbf{x} \\ &= \frac{\partial}{\partial\theta} \int T_n(\mathbf{x}) f(\mathbf{x}|\theta)\, \mathrm{d}\mathbf{x} \qquad \text{under regularity conditions} \\ &= \frac{\partial}{\partial\theta} \mathbb{E}[T_n] \\ &= \frac{\partial}{\partial\theta} \theta \qquad \text{using } T_n \text{ is unbiased} \\ &= 1 \end{aligned}$$

By the Cauchy-Schwarz inequality (see page 143), we find

$$\begin{aligned} 1 = \mathrm{Cov}(T_n, Z_n) &\leq \sqrt{\mathrm{Var}(T_n)\mathrm{Var}(Z_n)} \\ &= \sqrt{\mathrm{Var}(T_n)}\sqrt{nI(\theta)} \end{aligned}$$

and the result follows. ■

We can easily extend this result. Suppose $T_n$ is an unbiased estimator of $g(\theta)$ for some differentiable function $g : \mathbb{R} \to \mathbb{R}$. Then the above proof shows that $\mathbb{E}[Z_n T_n] = g'(\theta)$ and we arrive at

$$\mathrm{Var}(T_n) \geq \frac{(g'(\theta))^2}{nI(\theta)}$$

For the proof of the Cauchy-Schwarz inequality, this follows immediately from the fact that the correlation is between $[-1, 1]$. This in turn follows from

$$\mathrm{Var}\left(\frac{X}{\sigma_X} \pm \frac{Y}{\sigma_Y}\right) = 2 \pm 2\rho_{XY}.$$

There are also multivariate extensions ($\theta \in \mathbb{R}^d$), which we don't discuss here.

We end with a few examples.

(1) Poisson($\mu$). The MLE $\widehat{\mu} = \overline{X}_n$ is unbiased, and attains the Cramer-Rao lower bound as

$$\mathrm{Var}(\widehat{\mu}) = \frac{\mu}{n} = \frac{1}{nI(\mu)}$$

We say that $\widehat{\mu}$ is efficient, and conclude that there is no other unbiased estimator with a smaller variance.

(2) Normal($\mu, 1$). Again, the MLE is $\widehat{\mu} = \overline{X}_n$, which is unbiased and efficient as

$$\mathrm{Var}(\widehat{\mu}) = \frac{1}{n} = \frac{1}{nI(\mu)}$$

as $I(\mu) = 1$.

(3) Hardy-Weinberg (multinomial with $m = 3$ and $p_1 = (1-\theta)^2$, $p_2 = 2\theta(1-\theta)$, $p_3 = \theta^2$). We have seen that

$$\widehat{\theta} = \frac{X_2 + 2X_3}{2n}$$

is unbiased and has variance $\theta(1-\theta)/(2n)$, which coincides with $1/\{I(\theta)\}$.

(4) Exponential($\lambda$). Here the MLE $\widehat{\lambda} = 1/\overline{X}_n$ is biased as

$$\mathbb{E}[\widehat{\lambda}] = \frac{n}{n-1}\lambda$$

We can tweak it a little, and define

$$\widetilde{\lambda} := \frac{n-1}{n}\widehat{\lambda}$$

This estimate is unbiased, and has variance

$$\mathrm{Var}(\widetilde{\lambda}) = \frac{(n-1)^2}{n^2}\mathrm{Var}(\widehat{\lambda}) = \frac{(n-1)^2}{n^2}\frac{n^2\lambda^2}{(n-1)^2(n-2)} = \frac{\lambda^2}{n-2}$$

which is only slightly larger than

$$\frac{1}{nI(\lambda)} = \frac{\lambda^2}{n}$$

We cannot conclude that $\widehat{\lambda}$ or $\widetilde{\lambda}$ is efficient from the theorem.

## Uniqueness of the UMVUE

Suppose $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are two unbiased estimators, and both have minimal variance among all unbiased estimators. Then

$$\mathbb{P}\{\widehat{\theta}_1 = \widehat{\theta}_2\} = 1.$$

We can prove this as follows. Write $\sigma^2$ for the minimal variance. Consider

$$\overline{\theta} = \frac{1}{2}\widehat{\theta}_1 + \frac{1}{2}\widehat{\theta}_2$$

This estimator is again unbiased and

$$\sigma^2 \leq \mathrm{Var}(\overline{\theta}) = \frac{1}{2}\sigma^2 + \frac{1}{2}\mathrm{Cov}(\widehat{\theta}_1, \widehat{\theta}_2)$$

which implies

$$\mathrm{Cov}(\widehat{\theta}_1, \widehat{\theta}_2) \geq \sigma^2$$

By Cauchy-Schwarz, the inequality

$$\mathrm{Cov}(\widehat{\theta}_1, \widehat{\theta}_2) \leq \sigma^2$$

always holds, so we get

$$\mathrm{Cor}(\widehat{\theta}_1, \widehat{\theta}_2) = 1$$

which is possible only if

$$\widehat{\theta}_1 = a + b\widehat{\theta}_2$$

for some $a, b$. Since $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are unbiased, we find

$$\theta = a + b\theta \qquad \forall \theta,$$

so we must have $a = 0$ and $b = 1$, and the conclusion follows.

Easier? Using the Bienaymé -Chebyshev's inequality, we find

$$\begin{aligned}
\mathbb{P}\{|\widehat{\theta}_1 - \widehat{\theta}_2| \geq \varepsilon\} &\leq \frac{\mathbb{E}[|\widehat{\theta}_1 - \widehat{\theta}_2|^2]}{\varepsilon^2} \\
&= \frac{\mathrm{Var}(\widehat{\theta}_1 - \widehat{\theta}_2)}{\varepsilon^2} \qquad \text{since } \mathbb{E}[\widehat{\theta}_1] = \mathbb{E}[\widehat{\theta}_2] \\
&= \frac{2\sigma^2 - 2\mathrm{Cov}(\widehat{\theta}_1, \widehat{\theta}_2)}{\varepsilon^2} \\
&= 0
\end{aligned}$$

for all $\varepsilon > 0$, and the claim follows.